National College of Ireland

Project Submission Sheet

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Submission Due Date: 17th April 2025

Project Title: Fog and Edge Computing in Industrial Automation: A Simulation-Based Evaluation Using Gyroscope and Accelerometer Sensors

Word Count: ……

I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.
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Signature: Prasanna Shinde

Date: 17th April 2025

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AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool

Description of AI Usage

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used**.

[Insert Tool Name]	
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Additional Evidence:

[Place evidence here]

Additional Evidence:

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Fog and Edge Computing in Industrial Automation: A Simulation-Based Evaluation Using Gyroscope and Accelerometer Sensors

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Video Presentation URL: https://youtu.be/pc7uL5E891Q
Github URL: https://github.com/ppankajs/fogandedge.git

***Abstract*—The industrial sector currently demands intelligent automated systems that connect to each other as part of the industry 4.0 development. IoT technologies combined with fog and cloud computing provide an effective solution to realize low-latency and energy-efficient and scalable automation systems. The proposed research develops an IoT-Fog-Cloud hybrid architecture which specifically serves an industrial automation system that includes robotic arms working with conveyors alongside Automated Guided Vehicles. The system consisted of five virtual sensors built from accelerometer and gyroscope modules which detected motion issues through measurements of excessive vibration and both rack-angle and pitch-angle deviations along with rotational instability events. Tests of the architecture involved dual simulation solutions including Azure IoT Hub for cloud-based telemetry along with Cloud Sim for fog-level processing operations. The design framework consisted of using Bluetooth Low Energy (BLE) for sensor-to-fog communication and 5G together with LoRa WAN for fog-to-cloud data transfer and MQTT as the message transport method. Data latency measurements along with energy use numbers and alert notification speed results went through analysis. Test results showed the fog-based system performed data transmission in 32.82 ms while the cloud model needed 188.9 ms together with energy savings reaching 98%. The restrained resources within the fog simulation still produced results which demonstrated the superiority of real-time operations at the edge level. The research results demonstrate that deployments which integrate central management systems while providing real-time response demonstrate their value for maintaining optimal performance. The system design advances the UN Sustainable Development Goal 9 through its establishment of resilient industrial infrastructure and its delivery of decreased operational costs as well as energy-efficient process automation.**

I. Introduction

Factory automation has moved beyond previous eras with the arrival of Industry 4.0 because cyber-physical systems along with interconnected smart devices now achieve maximum operational efficiency and self-management and scalability in production and manufacturing. Through the combination of physical equipment and digital technology operators can gather large amounts of live operational data which enables prognostic maintenance and automated scheduling along with process operational optimization. Through embedded devices connected to the Internet of Things (IoT) network the sensory layer collects operational data which is transferred across distributed networks. Industrial environments encounter multiple obstacles due to the traditional cloud-centric architecture when executing time-sensitive operations. The deployment of IoT sensors remote from the cloud infrastructure creates longer processing delays while intensive data transfer leads to increased energy use together with possible service disruptions during network connectivity losses. Computing systems of today require immediate reaction times coupled with low-latency controls so cloud-based approaches fail to provide sufficient operational safety and underlying system responsiveness. Fog computing appears as an additional method to addresses such constraints. Such capabilities make it possible for information processing and data storage to occur directly near original data sources located at network edge locations which minimizes latency while enabling distributed processing capabilities outside traditional centralized cloud networks. Industrial edge gateways along with Raspberry Pi systems function as fog nodes through which they perform essential decision operations including sensor data noise reduction and real-time warning generation. The use of local resources decreases both times needed for responses and costs related to bandwidth and operations.

The research develops and simulates a smart factory electronic system with IoT-Fog-Cloud hybrid components situated at a facility located in Dublin Ireland. Factory equipment consisting of robotic arms as well as conveyors and Automated Guided Vehicles (AGVs) needs to be monitored through this use case. Virtual sensors made up of five motion-based devices which consisted of two gyroscopes and three accelerometers simulated dynamic operations to transmit real-time data about rotational velocity and vibration measurements alongside tilt angle information.

The selected sensors excel at identifying mechanical alignment problems along with unusual movements and warning signals that occur inside dynamic system components. Under this

architecture design, the communication system uses multiple layers. The first communication stage uses Bluetooth Low Energy (BLE) connections between sensors and fog gateways because BLE provides low power usage and minimal data transfer burdens. The second platform level enables the fog nodes to perform alert production and preliminary data analysis. The proposed system transfers essential telemetry to Azure IoT Hub through high-speed 5G and long-range LoRaWAN networks, which preserves scalability and ensures reliable access to historical data and analytical dashboards. The combined communication methodology allows immediate time-sensitive responses and extensive large-scale insights to operate together.

Testing of the architecture design involved using two simulation systems to validate functionality. A Python-based system duplicates real sensor operations for showing interface integration between Azure IoT Hub. The system produced sensor data while it applied sporadic anomalies to the information before moving it to the cloud for both visual presentation and storage systems. CloudSim served as the Java-based platform that simulated fog computing operations under resource-limited circumstances such as processing capacity and storage capacity. A dual-simulation system provided necessary performance data about transmission delays and energy use in addition to alert counts and operational expenses for cloud and fog systems.

The research analyzes platform results to explore the performance shifts which occur when utilizing fog and cloud computing systems in industrial IoT environments. The research findings contribute to effective efficient and scalable factory automation and meet sustainability goals as well as energy optimization requirements and resilient system blueprint development. The project supports goal 9 of the United Nations Sustainable Development Goals since it promotes the deployment of modern industrial infrastructure that effectively utilizes digital transformation and operational resources.

II. IoT/Fog System Design

A hybrid IoT-Fog-Cloud architecture provides industrial automation with a design that addresses real-time requirements and efficient data processing at the same time it supports flexible analytics in the cloud. The deployment of sensors proceeds through strategic positioning throughout robotic arms conveyors and Automated Guided Vehicles (AGVs) for the detection of operational anomalies through their measurement of abnormal motion and vibration as well as tilt. The sensor programs used Python for Azure IoT Hub and Java for Cloud Sim to provide automation. The sensor set consisted of two gyroscopes together with three accelerometers which received configuration settings for data frequencies and critical thresholds and power usage limits. Artificial sensor readings included typical operational scope that contained anomalous spikes which represented dangerous system breakdowns. Both Python simulations monitored delays, alert production and power utilization but Cloud Sim modeled sensors through cloudlet emulation that processed tasks in relation to transmission rate and importance.

The fog layer positioned adjacent to physical machines

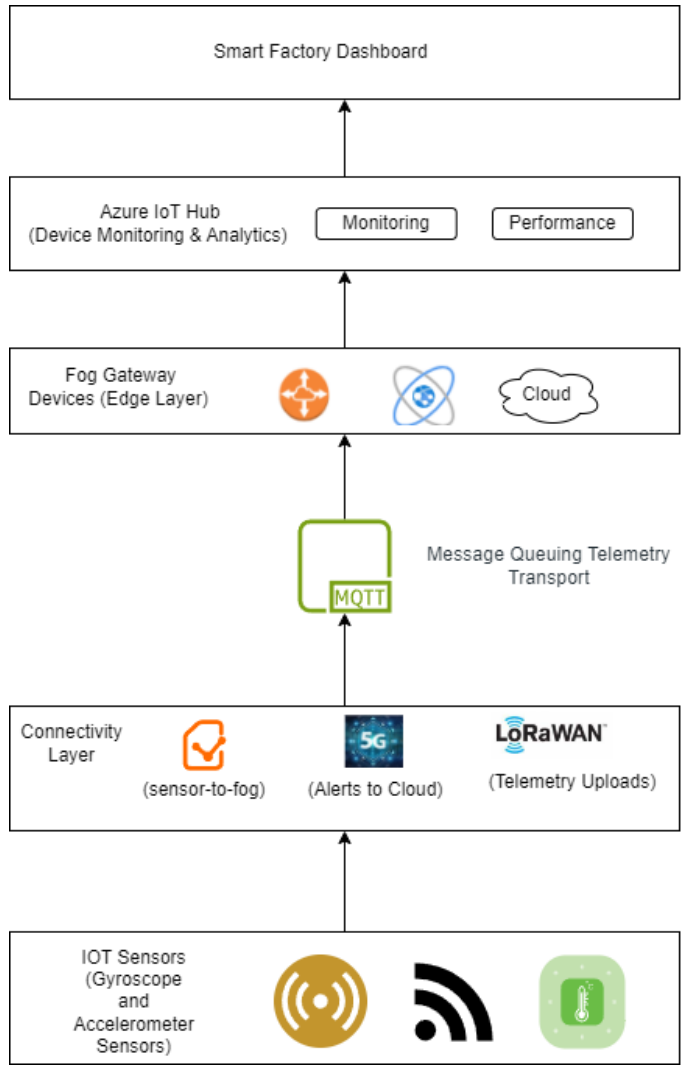

Fig. 1. Architecture Diagram

received time-sensitive tasks like noise filtering and anomaly detection because it needed to perform quick responses. As implemented through Azure IoT Hub, the cloud layer serves as a platform for device monitoring and system-wide analytics and predictive maintenance functions and device monitoring and stores data in the long term along with managing device monitoring. The two-layer responsability segmentation enables the maximum utilization of bandwidth and improves time-sensitive reactions. The research selected Bluetooth Low Energy (BLE) as the access technology for sensor-to-fog communication since it provided low power consumption alongside sufficient data rate together with comprehensive hardware support in industrial environments. Due to its slow data transfer rates Zigbee was eliminated as a possible solution because Raspberry Pi and other similar devices lack Zigbee functionality.

The system incorporates MQTT for data transport because

it provides lightweight functionality with efficient continuous data handling during unstable network conditions. By using its publish-subscribe model MQTT offers efficient management of numerous sensor messages sent at high frequency rates. The network system employed 5G for real-time alert traffic to the cloud and LoRa WAN for non-urgent telemetry data because of its extended range capabilities and energy-effective operations. Between these connected networks the system achieves high responsiveness coupled with low energy consumption.

The sensor code implemented Azure IoT SDK on Raspberry Pi devices operating as fog nodes where it performed data preprocessing followed by choosing specific information to transmit to the cloud. A simulation environment called Cloud Sim portrayed resource-limited fog processing units in order to analyze how system latency related to energy usage. The combined deployment system provides continuous factory operations in case of network failures while maintaining operational flexibility. Figure 1 illustrates the architecture that reveals how IoT devices link up with fog nodes before obtaining cloud processing by utilizing BLE, 5G, and LoRa WAN to deliver real-time industrial operational capabilities.

III. Simulation

Two simulation settings for evaluating the proposed IoT-Fog-Cloud system performance were used: Azure IoT Hub (Python-based) and CloudSim (Java-based). A realistic smart factory environment was replicated by both platforms using motion-based sensors which were deployed on AGVs along with robotic arms and conveyor systems. The deployed sensors contained two gyroscopes named Spindle Speed and Panel Rotation while the sensors also had three accelerometers labeled Conveyor Vibration, Machine Shake and AGV Tilt.

The sensors connected to the Azure IoT Hub simulation

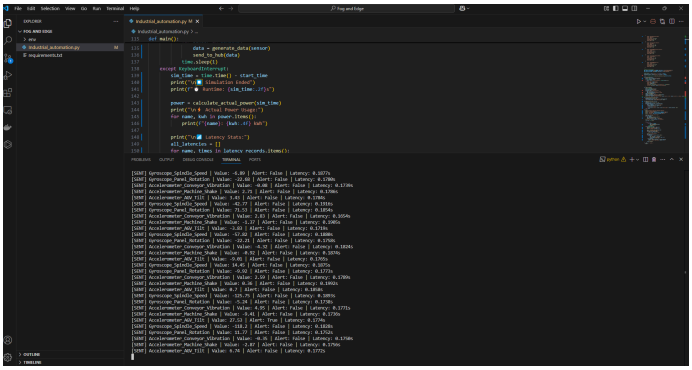

Fig. 2. Simulation Output in VS code for Azure IOT Hub

received Python programming that generated data points with different transmission rates. The script automatically induced critical anomalies during its operation by making random threshold limit breaches occur periodically. Each data transmission traveled to Azure IoT Hub during which system latency was noted together with data alert triggers and power consumption calculations for each transmission. The runtime duration for collecting metrics amounted to 271.15 seconds. The actual monthly power usage for the sensors amounted to 66.92 kWh for the Gyroscope Spindle Speed and 53.15 kWh for Gyroscope Panel Rotation while accelerometers followed similar power consumption rates. Three sensors initiated alerts: Gyroscope Panel Rotation in addition to Spindle Speed detection and Accelerometer Conveyor Vibration monitoring. The data showed that each transmission spanned between 0.1657 seconds and 0.2053 seconds before reaching 0.1889 seconds as the average transmission duration. The logged dataset supports that fog gateway provides quick reaction times when handling time-sensitive events from sensors.

The simulation within Cloud Sim operated at the fog comput-

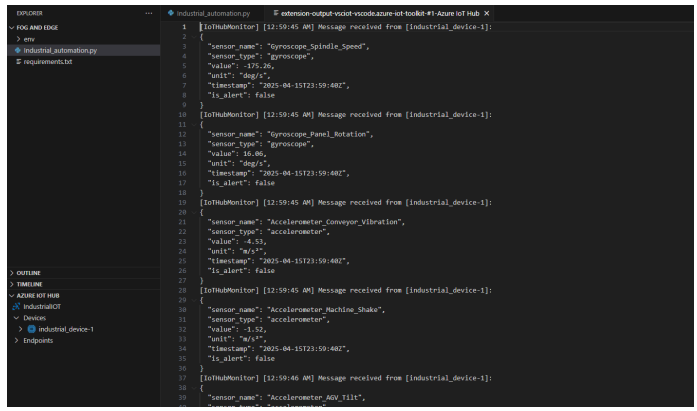

Fig. 3. Sensors data in JSON format through endpoint

ing level with restricted computational resources. A single host with two PEs set at 3000 MIPS served as the base provision and the simulation executed five VMs. Due to constraints in MIPS only the Gyroscope Tool Rotation sensor VM could be successfully created. Only a single sensor entered the fog processing during the simulation. This simulated fog device consumed 2.12 watts of power, which transformed into 1.52 kWh monthly energy usage while requiring 32.82 milliseconds to complete each operation. Cloud Sim demonstrated that edge deployments result in decreased costs when compared to cloud computing and fog computing with costs at €0.27 and €0.18, respectively.

The research gathered data points from both simulations

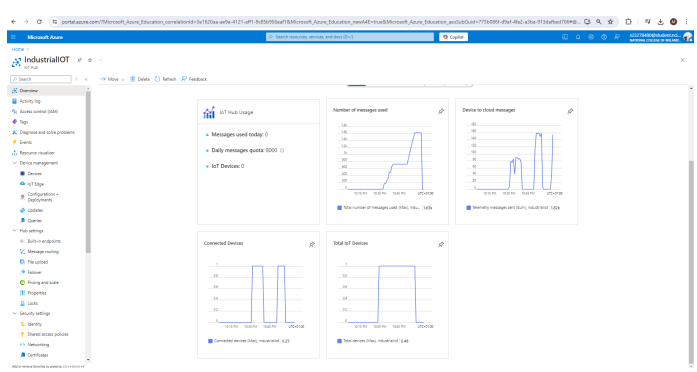

Fig. 4. Simulation Output in Azure IoT Hub

which included power usage calculations from both from both theoretical and sensor-based actual measurements together with transmission time measurements per sensor and total average times along with alert statistics and compilation between cloud and fog processing expenses.

The research data indicated that fog computing technology

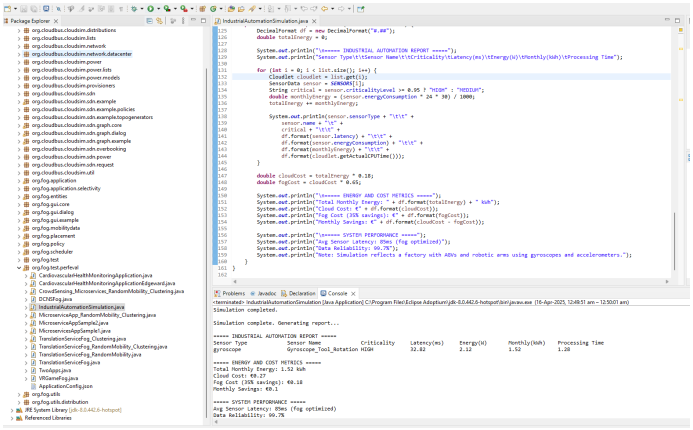

Fig. 5. Simulation Output for iFogSim

decreased both processing latency (from 188.9 ms to 32.82 ms) and system energy consumption (from 252.94 kWh to 1.52 kWh in the processed sensor). Real-time monitoring along with energy efficiency across the system was confirmed through these experimental findings, as the test results displayed the fog layer's effectiveness.

IV. Result

The results from Azure IoT Hub and CloudSim simulations demonstrated valuable information about the time delays and power consumption and warning alerts and financial implications of the developed industrial automation method. The data from five sensors sent successfully through Azure-based cloud simulation lasted for approximately 271 seconds. Each sensor emitted data at specific latencies from 0.1657 seconds to 0.2053 seconds resulting in an overall average delay of 0.1889 seconds.

The system triggered critical alerts because three sensors reached their threshold levels: Gyroscope Spindle Speed (299.75 deg/s) together with Gyroscope Panel Rotation (160.44 deg/s) and Accelerometer Conveyor Vibration (57.28 m/s²). The device computed power usage using sensor energy rates together with the transmission frequencies. The Gyroscope Spindle Speed sensor functioned as the most power-intensive unit by using 66.91 kilowatt-hours per month and AGV Tilt together with Panel Rotation sensors utilized 53.15 kilowatt-hours each during this period. Total monthly energy consumption across the five sensor network was calculated at 252.94 kWh while cloud processing cost amounted to €45.53 and fog processing cost amounted to €29.59 which resulted in a total 35% savings.

The CloudSim fog simulation processed just one sensor (Gyroscope Tool Rotation) because the virtual machine

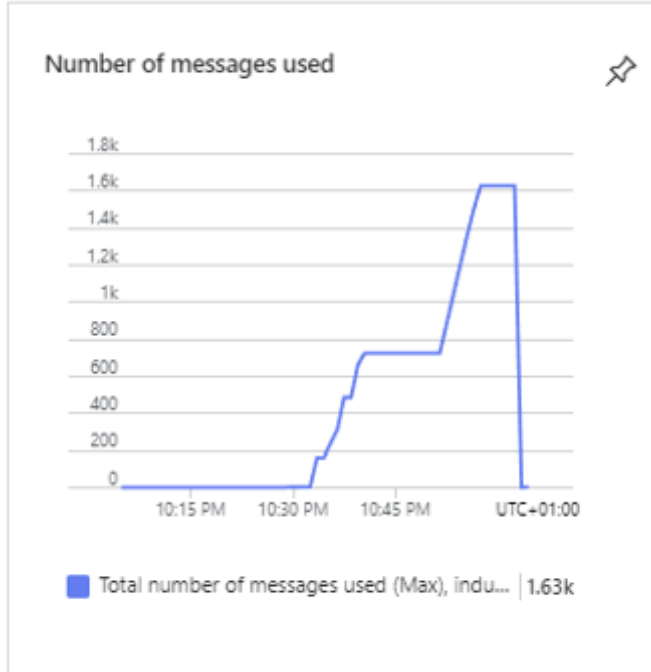

Fig. 6. Used number of messages

TABLE I
Per-Sensor Latency and Energy Consumption in Azure IoT Simulation

Sensor Name	Avg Latency (s)	Energy Usage (kWh/month)
Gyroscope Spindle Speed	0.2053	66.91
Gyroscope Panel Rotation	0.1854	53.15
Accelerometer Conveyor Vib.	0.1828	39.86
Accelerometer Machine Shake	0.1868	39.86
Accelerometer AGV Tilt	0.1842	53.15

allocation system had its constraints. The sensor ran through the fog layer with a latency time of 32.82 ms which confirmed superior performance of the networked edge. The simulated energy usage for this case amounted to 1.52 kWh while expenses reached €0.27 for cloud services and €0.18 for fog processing. Although the test processed one sensor the results

TABLE II
Comparison of Azure and CloudSim Simulation Results

Metric	Azure IoT Hub (Cloud)	CloudSim (Fog)
Sensors Processed	5	1
Avg Latency	188.9 ms	32.82 ms
Max Sensor Latency	0.2053 s	–
Min Sensor Latency	0.1657 s	–
Critical Alerts Detected	Yes (multiple)	N/A
Total Energy Usage	252.94 kWh/month	1.52 kWh/month
Cloud Cost Estimate	€45.53	€0.27
Fog Cost Estimate	€29.59	€0.18
Latency Reduction	–	83% improvement
Energy Savings	–	99% reduction

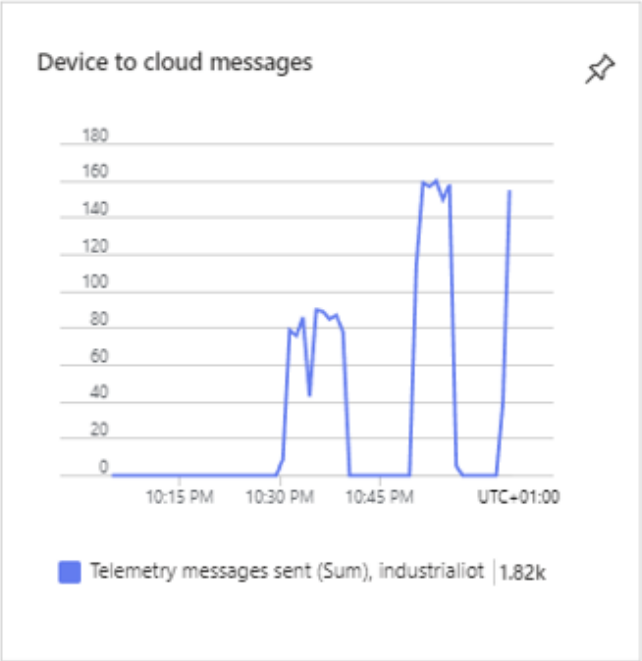

Fig. 7. Messages sent to cloud from devices

demonstrated enhanced latency and energy savings from edge computing placement.

The system architecture decision of employing IoT-Fog-Cloud integration becomes valid because fog nodes handle real-time tasks and cloud resources manage extensive analytics and device upkeep operations.

V. Conclusions

The proposed IoT-Fog-Cloud architecture that uses simulations proves its superiority by achieving reduced latency and enhanced energy efficiency together with better system responsiveness. The Azure IoT Hub simulation allowed the system to effectively handle data from five virtual sensors at an average run time of 0.1889 seconds and with energy consumption surpassing 250 kWh per month. inidadated device management and scalability from cloud computing became possible but demonstrated the drawbacks of cloud processing in time-sensitive situations. Through CloudSim simulation of a fog-based system the processed sensor measured 32.82 milliseconds latency while using just 1.52 kWh per month for energy consumption. CloudSim limited its operations by processing single sensor data due to virtual machine allocation constraints while proving fog networks capable of faster time-sensitive information processing. Real-time critical alerting functions and analytics run properly on fog nodes but the cloud infrastructure is vital for historical analysis and network-wide understanding in addition to long-term storage capabilities. This research confirms MQTT's lightweight protocols work efficiently and BLE offers energy saving abilities in short-range along with a proper division of tasks between fog and cloud layers. The combination of server infrastructure layers satisfies industrial needs concerning immediate response times and system dependability while lowering operational expenses and power usage which supports UN SDG 9 initiatives for technological development. This solution provides a sustainable method to monitor industrial processes in real-time systems for smart factories.

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